Prakhar Kumar

Bachelor of Technology
Computer Science and Engineering
National Institute of Technology, Patna

+91-9934365253
prakhark.ug24.cs@nitp.ac.in
linkedin.com/in/Prakhar Kumar
github.com/Prakhar-k

Education

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech., CSE	National Institute of Technology, Patna	8.43,	June 2026
Gyanodaya Gurukul	CBSE	85.2%	May 2023
ST.Xaviers's English Medium School	CBSE	93.83%	May 2021

Projects

- **Project A: Full Stack E-Commerce Application** *2026*
  Frameworks: React, Spring Boot *Tools: VS Code, Git, GitHub, IntelliJ IDEA* [|]
  - Developed a full-stack e-commerce application using React, Tailwind CSS, Spring Boot, and MySQL.
  - Implemented role-based authentication with separate Admin and Seller panels for authorized users.
  - Built product, seller, order management, user authorization, and role-specific functionality for customers, sellers, and administrators.

- **Project B: Paimana Project Management Platform** *2026*
  Frameworks: Spring Boot, FastAPI *Tools: PostgreSQL, Git, GitHub, IntelliJ IDEA* GitHub
  - Developed an AI-driven project monitoring and decision-support platform to track infrastructure projects, assess risks, and forecast financial overruns.
  - Built secure REST APIs using Spring Boot, Spring Security, JWT Authentication, and PostgreSQL for project, sector, and monthly progress management.
  - Designed and integrated a FastAPI-based machine learning microservice utilizing Random Forest and Quantile Regression models for risk prediction and overrun forecasting.
  - Implemented automated feature engineering and SHAP-based Explainable AI to provide transparent, feature-level insights into risk predictions.
  - Integrated Google Gemini through Spring AI to generate executive decision briefs and actionable recommendations from project analytics.
  - Engineered a fault-tolerant fallback mechanism that ensures uninterrupted prediction services when the ML microservice is unavailable.

Skills

- **Programming Languages**: C, C++, Java, Python, HTML, CSS, JavaScript , SQL
- **Frameworks**: React, Spring Boot , FastAPI
- **Libraries**: NumPy, Pandas, Matplotlib, Seaborn , SHAP
- **Tools**: VS Code, IntelliJ IDEA, Git, GitHub, Render , Postman
- **Core CS Subjects**: Operating Systems, DBMS, Computer Networks, Computer Organization and Architecture
- **Areas of Expertise**: Data Structures & Algorithms (DSA), OOP's, Web Development, Artificial Intelligence (AI) , Machine Learning (ML)

Certifications

- **HackSlash , NITP**, ByteVerse Hackathon *2025*
- **NPTEL**, NPTEL Certificate *2026*

Achievements

- **DSA** 500+ Data Structures and Algorithms questions solved on LeetCode
  - Peak Contest Rating: 1576
  - Cleared Smart India Hackathon internal round
- **JEE Main** 98.7 percentile